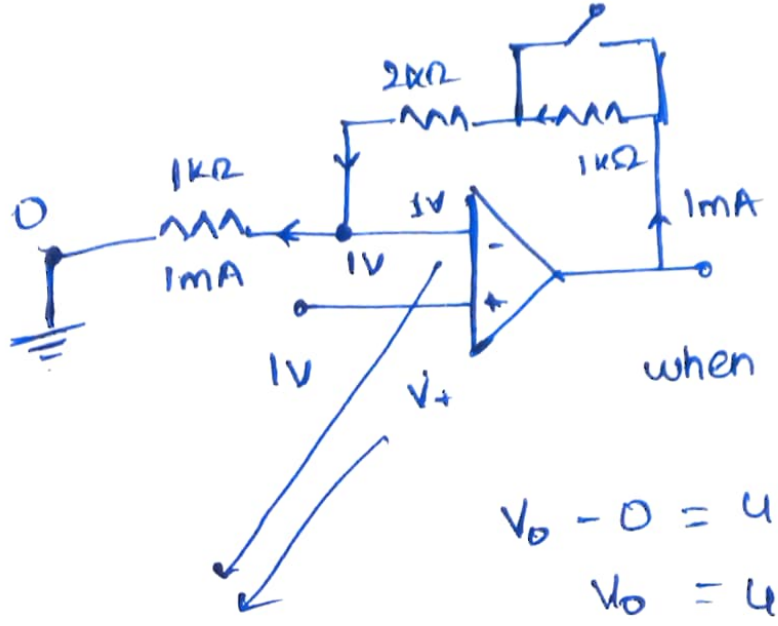


when S is open

$$V_o - 0 = 4k\Omega\,(1mA)$$

$$V_o = 4V$$

Same voltage.

when S is closed

$$V_o' - 0 = 3k\Omega\,(1mA)$$

$$= 3V$$

$$\therefore V_o' = \frac{3}{4} V_o = \frac{3x}{4}.$$